

Qualification Pack



Autonomous Mobile Robotics

QP Code: ELE/Q6704

Version: 2.0

NSQF Level: 4.5

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ELE/Q6704: Autonomous Mobile Robotics

Brief Job Description

A Robotics Expert is a skilled professional involved in the design, development, maintenance, and implementation of robotic systems, including Autonomous Mobile Robotics (AMR), across diverse industries. They operate in various environments such as offices, manufacturing plants, and research laboratories focusing on solving sector specific technical and operational challenges. In manufacturing, for example, robotics technologists may develop robotic automation solutions to enhance productivity or design autonomous mobile robots for tasks in hazardous, complex, or hard to reach areas. The role requires strong expertise in robotics fundamentals, including logic design, microprocessors, programming, and mechanical, electrical, and control systems. These professionals work collaboratively with multidisciplinary teams to build efficient, reliable, and high performing robotic and AMR solutions.

Personal Attributes

In manufacturing, robotics technologists may create robots to automate tasks and increase productivity, or they might design mobile robots for exploration in dangerous or hard-to-reach areas. The role requires a strong understanding of robotics, including logic, microprocessors, computer programming, and mechanical, electrical, and control systems. Robotics technologists collaborate with multiple disciplines to design and build the most efficient and effective robots.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ELE/N6708: Work organization and management \(Autonomous Mobile Robotics\)](#)
2. [ELE/N6709: Communication and interpersonal skills \(Autonomous Mobile Robotics\)](#)
3. [ELE/N6707: Design \(Autonomous Mobile Robotics\)](#)
4. [ELE/N6711: Navigation and localization \(Autonomous Mobile Robotics\)](#)
5. [ELE/N6710: Prototyping \(Autonomous Mobile Robotics\)](#)
6. [ELE/N6712: Vision \(Autonomous Mobile Robotics\)](#)
7. [ELE/N6713: Robot environment interaction \(Autonomous Mobile Robotics\)](#)
8. [ELE/N6714: Testing and fault finding \(Autonomous Mobile Robotics\)](#)
9. [ELE/N6715: Performance Review and Commissioning \(Autonomous Mobile Robotics\)](#)

Qualification Pack (QP) Parameters

Qualification Pack

Sector	Electronics
Sub-Sector	Industrial Automation
Occupation	Product Design & Development-EM&B
Country	India
NSQF Level	4.5
Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/3139.1400
Minimum Educational Qualification & Experience	No formal education prescribed
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	14 Years
Last Reviewed On	NA
Next Review Date	06/02/2028
NSQC Approval Date	06/02/2026
Version	2.0
Reference code on NQR	QG-4.5-EH-05203-2026-V1-ESSCI
NQR Version	2

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ELE/N6708: Work organization and management (Autonomous Mobile Robotics)

Description

This NOS unit is about proper training, the correct use of equipment, and adherence to safety protocols in manufacturing settings.

Scope

The scope covers the following :

- Principles and Applications of Safe Working
- Purposes, Uses, Care, and Maintenance of Equipment and Materials
- Environmental and Safety Principles in Good Housekeeping
- Principles of Team Working and Their Applications

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Maintain a clean, safe, and well-organized work area in accordance with workplace housekeeping and environmental guidelines.
- PC2.** Prepare self appropriately for assigned tasks, ensuring full compliance with health, safety, and personal protective requirements.
- PC3.** Schedule work activities efficiently to optimize productivity and minimize disruption to ongoing operations.
- PC4.** Follow all rules, regulations, and standard operating procedures applicable to robotics technician/engineering tasks.
- PC5.** Select appropriate tools, equipment, and materials required for the task.
- PC6.** Use all tools, equipment, and materials safely and strictly as per manufacturer's instructions.
- PC7.** Demonstrate adherence to safety standards for the environment, equipment, materials, and work processes.
- PC8.** Restore the work area to its proper condition after completing tasks, ensuring safe and ready-to-use status.
- PC9.** Contribute effectively to team tasks by providing support, receiving and giving feedback, and demonstrating collaborative behaviour.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of mobile robot design and development processes.
- KU2.** Familiarity with the diverse applications of mobile robotics in various industries like autonomous vehicles, manufacturing, agriculture, aerospace, mining, and medicine.

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- KU3.** Awareness of the challenges posed by environments that are unsafe or inaccessible to humans, such as hazardous areas or hard-to-reach places.
- KU4.** Proficiency in translating client requirements into accurate and effective specifications.
- KU5.** Knowledge of how to program, test, and evaluate the robots to ensure they perform optimally in real-world conditions.
- KU6.** Competence in utilizing microprocessors for robot operation and control.
- KU7.** Ability to design robots in a cost-efficient manner, considering factors like material costs and production time.
- KU8.** Strong interpersonal and communication skills, with the ability to work well in a team of engineers, designers, and other professionals.
- KU9.** Understanding the global nature of the robotics industry and the importance of working across different cultures and regions.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to analyze complex problems and develop effective, practical solutions, especially in unique and challenging environments.
- GS2.** Ensuring precision in design, development, testing, and troubleshooting for high-performing robots.
- GS3.** Flexibility to work across diverse sectors like manufacturing, agriculture, mining, aerospace, and medicine, and adjust to fast-paced technological advancements.
- GS4.** Strong interpersonal skills and the ability to work in multidisciplinary teams with diverse skill sets to create and refine robotic systems.
- GS5.** The ability to work in international environments, dealing with diverse cultures and understanding their needs and expectations.
- GS6.** Proficiency in key technologies, including microprocessors, computer programming, and control systems (mechanical, electrical).
- GS7.** Ability to design, build, and prototype robots that meet client needs, ensuring functionality, safety, and performance.
- GS8.** Understanding cost constraints and providing efficient solutions that meet both the budgetary and technical requirements of the project.
- GS9.** Managing the entire lifecycle of robotic projects, from consultation and design to deployment and maintenance, while ensuring deadlines and objectives are met.
- GS10.** Conducting careful client consultations to understand needs, and translating them into detailed specifications for robotic applications.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	10	-	-
PC1. Maintain a clean, safe, and well-organized work area in accordance with workplace housekeeping and environmental guidelines.	-	2	-	-
PC2. Prepare self appropriately for assigned tasks, ensuring full compliance with health, safety, and personal protective requirements.	-	1	-	-
PC3. Schedule work activities efficiently to optimize productivity and minimize disruption to ongoing operations.	-	1	-	-
PC4. Follow all rules, regulations, and standard operating procedures applicable to robotics technician/engineering tasks.	-	1	-	-
PC5. Select appropriate tools, equipment, and materials required for the task.	-	1	-	-
PC6. Use all tools, equipment, and materials safely and strictly as per manufacturer's instructions.	-	1	-	-
PC7. Demonstrate adherence to safety standards for the environment, equipment, materials, and work processes.	-	1	-	-
PC8. Restore the work area to its proper condition after completing tasks, ensuring safe and ready-to-use status.	-	1	-	-
PC9. Contribute effectively to team tasks by providing support, receiving and giving feedback, and demonstrating collaborative behaviour.	-	1	-	-
NOS Total	-	10	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6708
NOS Name	Work organization and management (Autonomous Mobile Robotics)
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	2
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

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ELE/N6709: Communication and interpersonal skills (Autonomous Mobile Robotics)

Description

This NOS unit is about the communication and interpersonal skills.

Scope

The scope covers the following :

- Range and Purposes of Documentation and Publications in Electronic Forms
- Technical Language Associated with the Skill and Technology
- Standards for Routine and Exception Reporting in Oral and Electronic Form
- Required Standards for Communicating with Clients, Team Members, and Others

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Identify and understand various types of documentation and publications available in electronic formats.
- PC2.** Interpret technical terminology and language relevant to the robotics and engineering domain.
- PC3.** Follow organizational standards for routine and exception reporting in oral, written, and electronic forms.
- PC4.** Communicate effectively with clients, team members, and stakeholders using appropriate professional standards.
- PC5.** Apply correct procedures for maintaining records, including technical, operational, and financial documentation.
- PC6.** Read, interpret, and extract relevant information from technical documents, manuals, and electronic data sources.
- PC7.** Conduct research to support problem-solving and continuous professional development.
- PC8.** Use oral, written, and electronic communication tools to convey information clearly and efficiently.
- PC9.** Operate a standard range of communication technologies such as email, digital documentation tools, and remote collaboration platforms.
- PC10.** Discuss complex technical concepts with peers and colleagues using appropriate technical language.
- PC11.** Explain technical principles and applications to non-experts in a simplified and understandable manner.
- PC12.** Prepare structured reports, documentation, and records while addressing client queries and ensuring compliance with business and safety guidelines.

Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** Knowledge of documentation formats
- KU2.** A strong understanding of technical terminology and concepts related to the field.
- KU3.** Ability to assess, analyze, and extract relevant data from documentation and convert that into actionable insights or instructions.
- KU4.** Ability to access and utilize resources like academic journals, technical papers, standards, and online databases to gather information for solving complex issues.
- KU5.** Apply critical thinking and structured approaches (e.g., root cause analysis, hypothesis testing) to resolve technical problems.
- KU6.** Ability to write detailed, well-structured reports, documentation, and instructions that are both comprehensive and accessible.
- KU7.** Proficiency in common office software (e.g., MS Word, Excel, PowerPoint), collaboration tools (e.g., Google Drive, Slack), and communication platforms (e.g., Zoom, Skype).
- KU8.** Knowledge of secure file-sharing systems (e.g., Dropbox, SharePoint) and collaborative project management platforms (e.g., Jira, Asana).
- KU9.** Ability to engage in in-depth discussions with colleagues or stakeholders on technical topics, offering explanations, insights, and alternative solutions.
- KU10.** Ability to translate highly technical concepts into everyday language for audiences without technical backgrounds.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to read, interpret, and extract technical data and instructions from documentation in various formats.
- GS2.** Utilize research techniques for solving complex problems and engaging in continuous professional development.
- GS3.** Communicate effectively through oral, written, and electronic means, ensuring clarity and efficiency.
- GS4.** Capability to discuss complex technical principles and applications with colleagues or team members.
- GS5.** Ability to explain complex technical concepts in simple terms to non-experts for better understanding.
- GS6.** Plan and arrange information gathering and preparation of documentation as required by the client.
- GS7.** Prepare and complete detailed reports, responding to arising issues and questions.
- GS8.** Respond to clients' needs both in person and indirectly (via email, phone, etc.).
- GS9.** Proficiency in using a standard range of communication technologies to facilitate communication.
- GS10.** Record decisions based on business principles, factoring in health, safety, and other essential considerations.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	10	-	-
PC1. Identify and understand various types of documentation and publications available in electronic formats.	-	1	-	-
PC2. Interpret technical terminology and language relevant to the robotics and engineering domain.	-	1	-	-
PC3. Follow organizational standards for routine and exception reporting in oral, written, and electronic forms.	-	1	-	-
PC4. Communicate effectively with clients, team members, and stakeholders using appropriate professional standards.	-	-	-	-
PC5. Apply correct procedures for maintaining records, including technical, operational, and financial documentation.	-	1	-	-
PC6. Read, interpret, and extract relevant information from technical documents, manuals, and electronic data sources.	-	-	-	-
PC7. Conduct research to support problem-solving and continuous professional development.	-	1	-	-
PC8. Use oral, written, and electronic communication tools to convey information clearly and efficiently.	-	1	-	-
PC9. Operate a standard range of communication technologies such as email, digital documentation tools, and remote collaboration platforms.	-	1	-	-
PC10. Discuss complex technical concepts with peers and colleagues using appropriate technical language.	-	1	-	-
PC11. Explain technical principles and applications to non-experts in a simplified and understandable manner.	-	1	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Prepare structured reports, documentation, and records while addressing client queries and ensuring compliance with business and safety guidelines.	-	1	-	-
NOS Total	-	10	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6709
NOS Name	Communication and interpersonal skills (Autonomous Mobile Robotics)
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	2
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

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ELE/N6707: Design (Autonomous Mobile Robotics)

Description

This NOS unit is about the principles and applications of project design, including the process of creating detailed project specifications and evaluating manufactured items based on specific criteria.

Scope

The scope covers the following :

- Principles and Applications of Project Design
- Nature and Formats of Project Specifications
- Bases for Appraisal of Manufactured Items
- Design Parameters

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Analyse project briefs and specifications to determine the required performance, functionality, and operational characteristics of mobile robotic systems.
- PC2.** Identify and resolve uncertainties, ambiguities, or missing information within project briefs or specifications.
- PC3.** Assess environmental conditions, operating constraints, and usage scenarios for which the mobile robot is being designed.
- PC4.** Identify and define hardware requirements—including mechanical, electrical, electronic, and control components—needed to support the robot's performance.
- PC5.** Generate design concepts and develop detailed project designs for manufacturing a functional robotic system within specified timelines.
- PC6.** Design tele-operation control systems independent of the robot's base platform, ensuring compatibility and performance.
- PC7.** Develop strategies for solving mobile robotics tasks such as navigation, obstacle avoidance, localisation, orientation, and mission execution.
- PC8.** Evaluate and select optimal options for materials, components, add-ons, and manufacturing processes based on cost, purpose, feasibility, and performance.
- PC9.** Complete the design and documentation process—including drawings, specifications, and standards—within defined cost, purpose, and time constraints.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding the concept of project design and its phases
- KU2.** Application of design principles to various project types
- KU3.** Integration of functional, aesthetic, and practical considerations in design

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- KU4.** Different formats of specifications (e.g., technical, functional, operational)
- KU5.** Standards for product assessment, including compliance and testing
- KU6.** Design Parameters
- KU7.** Principles and Applications for Designing, Assembling, and Commissioning Mobile Robotics Systems
- KU8.** Components and Functions of Electrical and Electronic Systems
- KU9.** Components and Applications of Add-Ons
- KU10.** Understanding the documentation requirements

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to understand and apply foundational principles in project design.
- GS2.** Knowledge of project lifecycle, including concept, planning, design, implementation, and commissioning.
- GS3.** Familiarity with the nature, structure, and formats of project specifications.
- GS4.** Understanding the criteria and bases for evaluating manufactured products (quality, cost, functionality, etc.).
- GS5.** Ability to evaluate various design options and make informed decisions based on criteria such as cost, functionality, and feasibility.
- GS6.** Understanding of principles and methods for organizing, controlling, and managing work within the context of a product's development.
- GS7.** Understanding the principles and applications involved in designing, building, and commissioning robotic systems for mobility.
- GS8.** Knowledge of the components and functions of electrical and electronic systems.
- GS9.** Understanding the components and applications of add-ons for mobile robotics (e.g., sensors, cameras, GPS modules).
- GS10.** Knowledge of the design and assembly principles for mechanical, electrical, and electronic systems.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	10	-	-
PC1. Analyse project briefs and specifications to determine the required performance, functionality, and operational characteristics of mobile robotic systems.	-	2	-	-
PC2. Identify and resolve uncertainties, ambiguities, or missing information within project briefs or specifications.	-	1	-	-
PC3. Assess environmental conditions, operating constraints, and usage scenarios for which the mobile robot is being designed.	-	1	-	-
PC4. Identify and define hardware requirements—including mechanical, electrical, electronic, and control components—needed to support the robot's performance.	-	1	-	-
PC5. Generate design concepts and develop detailed project designs for manufacturing a functional robotic system within specified timelines.	-	1	-	-
PC6. Design tele-operation control systems independent of the robot's base platform, ensuring compatibility and performance.	-	1	-	-
PC7. Develop strategies for solving mobile robotics tasks such as navigation, obstacle avoidance, localisation, orientation, and mission execution.	-	1	-	-
PC8. Evaluate and select optimal options for materials, components, add-ons, and manufacturing processes based on cost, purpose, feasibility, and performance.	-	1	-	-
PC9. Complete the design and documentation process—including drawings, specifications, and standards—within defined cost, purpose, and time constraints.	-	1	-	-
NOS Total	-	10	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6707
NOS Name	Design (Autonomous Mobile Robotics)
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	2
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

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ELE/N6711: Navigation and localization (Autonomous Mobile Robotics)

Description

This NOS unit is about navigation and localization of Robotics

Scope

The scope covers the following :

- Implement Navigation Strategies
- Develop Sensor-Based Environment Detection
- Implement and Test Map-Based Navigation
- Implement and Test Map Generation

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Implement navigation strategies for mobile robots based on orientation, mapping, and environmental conditions.
- PC2.** Develop and use sensor-based methods to detect, interpret, and understand the robot's surroundings.
- PC3.** Implement and test the robots ability to navigate using a pre-defined map of the environment.
- PC4.** Implement and test the robot's capability to autonomously generate maps of unknown environments.
- PC5.** Ensure accurate robot movement by applying orientation, localization, and mapping techniques.
- PC6.** Use industry-standard programming software to achieve reliable autonomous control of the robot's motion.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of navigation strategies based on orientation, mapping, and environmental conditions.
- KU2.** Knowledge of navigation strategies based on orientation, mapping, and environmental conditions.
- KU3.** Understanding of sensor-based perception and environmental interpretation methods.
- KU4.** Knowledge of map-based navigation using predefined environments.
- KU5.** Understanding of autonomous mapping techniques for unknown environments (e.g., SLAM).

Generic Skills (GS)

User/individual on the job needs to know how to:

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- GS1.** Knowledge of localization, orientation, and motion control principles.
- GS2.** Ability to implement navigation strategies for efficient robot movement.
- GS3.** Skill in using sensors to detect and interpret environmental data.
- GS4.** Ability to program and test navigation using predefined maps.
- GS5.** Ability to ensure accurate movement using localization and orientation techniques.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	8	-	-
PC1. Implement navigation strategies for mobile robots based on orientation, mapping, and environmental conditions.	-	2	-	-
PC2. Develop and use sensor-based methods to detect, interpret, and understand the robot's surroundings.	-	2	-	-
PC3. Implement and test the robots ability to navigate using a pre-defined map of the environment.	-	1	-	-
PC4. Implement and test the robot's capability to autonomously generate maps of unknown environments.	-	1	-	-
PC5. Ensure accurate robot movement by applying orientation, localization, and mapping techniques.	-	1	-	-
PC6. Use industry-standard programming software to achieve reliable autonomous control of the robot's motion.	-	1	-	-
NOS Total	-	8	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6711
NOS Name	Navigation and localization (Autonomous Mobile Robotics)
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	1
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

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ELE/N6710: Prototyping (Autonomous Mobile Robotics)

Description

This NOS unit is about Fabrication of Frame Parts

Scope

The scope covers the following :

- Design and manufacture the structural frame components of mobile robots
- Assemble and integrate the mechanical components, ensuring structural integrity and functionality
- Set up mechanical systems, electrical components, and sensors, ensuring all systems work in sync
- Integration of Structural and Mechanical Parts
- Integration of Electronic Control Circuits
- Physical and Software Adjustments

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Fabricate frame components of mobile robots using appropriate mechanical and fabrication techniques.
- PC2.** Assemble and integrate structural and mechanical parts of the mobile robotic system.
- PC3.** Integrate electronic control circuits, wiring, and related modules as per system design.
- PC4.** Install, configure, and perform necessary physical and software adjustments to ensure proper system functionality.
- PC5.** Set up and fine-tune mechanical, electrical, and sensor subsystems for reliable operation.
- PC6.** Perform efficient repair, replacement, and troubleshooting of faulty components.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of mechanical fabrication techniques for mobile robot frame construction.
- KU2.** Knowledge of mechanical fabrication techniques for mobile robot frame construction.
- KU3.** Understanding of assembly and integration of structural and mechanical components.
- KU4.** Understanding of system installation, configuration, and adjustment procedures.
- KU5.** Understanding of system installation, configuration, and adjustment procedures.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to set up and fine-tune subsystems for reliable performance.
- GS2.** Ability to fabricate robot frame components using appropriate techniques.

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- GS3.** Skill in assembling and integrating mechanical and structural parts accurately.
- GS4.** Ability to integrate electronic circuits and wiring as per design.
- GS5.** Skill in installing, configuring, and adjusting systems for proper functionality.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	10	-	-
PC1. Fabricate frame components of mobile robots using appropriate mechanical and fabrication techniques.	-	2	-	-
PC2. Assemble and integrate structural and mechanical parts of the mobile robotic system.	-	2	-	-
PC3. Integrate electronic control circuits, wiring, and related modules as per system design.	-	2	-	-
PC4. Install, configure, and perform necessary physical and software adjustments to ensure proper system functionality.	-	2	-	-
PC5. Set up and fine-tune mechanical, electrical, and sensor subsystems for reliable operation.	-	1	-	-
PC6. Perform efficient repair, replacement, and troubleshooting of faulty components.	-	1	-	-
NOS Total	-	10	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6710
NOS Name	Prototyping (Autonomous Mobile Robotics)
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	2
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

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ELE/N6712: Vision (Autonomous Mobile Robotics)

Description

This NOS unit is about knowledge of the application and functionality of different camera types, such as infrared (IR), depth cameras, and Lidar.

Scope

The scope covers the following :

- Impact of Light Conditions on Visualization Sensors
- Methods for Recognizing Color, Patterns, Objects, and Object Orientation
- Application of Different Camera Types (e.g., IR, Depth-Camera, Lidar):

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Install and configure cameras on mobile robots, ensuring appropriate positioning, calibration, and alignment.
- PC2.** Adjust camera settings to account for varying light conditions and ensure optimal visualization performance.
- PC3.** Implement vision systems that enable the robot to detect its environment and identify task-specific elements.
- PC4.** Apply image processing and object recognition algorithms to detect colours, shapes, patterns, and object orientation.
- PC5.** Integrate and test different types of cameras (IR, depth, LiDAR-based vision) for enhanced perception capabilities.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of camera installation, positioning, calibration, and alignment in robotic systems.
- KU2.** Understanding of camera settings and adjustments for varying lighting conditions.
- KU3.** Understanding of camera settings and adjustments for varying lighting conditions.
- KU4.** Understanding of image processing and object recognition techniques.
- KU5.** Understanding of image processing and object recognition techniques.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to install and configure cameras accurately on mobile robots.
- GS2.** Ability to install and configure cameras accurately on mobile robots.

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- GS3.** Ability to implement vision systems for environment detection and task execution.
- GS4.** Skill in applying image processing and object recognition algorithms.
- GS5.** Ability to integrate and test multiple camera systems for enhanced perception.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	8	-	-
PC1. Install and configure cameras on mobile robots, ensuring appropriate positioning, calibration, and alignment.	-	2	-	-
PC2. Adjust camera settings to account for varying light conditions and ensure optimal visualization performance.	-	2	-	-
PC3. Implement vision systems that enable the robot to detect its environment and identify task-specific elements.	-	2	-	-
PC4. Apply image processing and object recognition algorithms to detect colours, shapes, patterns, and object orientation.	-	1	-	-
PC5. Integrate and test different types of cameras (IR, depth, LiDAR-based vision) for enhanced perception capabilities.	-	1	-	-
NOS Total	-	8	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6712
NOS Name	Vision (Autonomous Mobile Robotics)
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	1
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

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ELE/N6713: Robot environment interaction (Autonomous Mobile Robotics)

Description

This NOS is about installing and adjusting sensors to ensure accurate environmental interaction.

Scope

The scope covers the following :

- Implement Actuators (Hardware and Software)
- Use Tele-operation for Robot Control
- Install and Adjust Sensors
- Use Manufacturer-Control Software for Autonomous Operation

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Implement and integrate actuators—both hardware and software—to enable robot interaction with the environment.
- PC2.** Apply tele operation techniques to achieve reliable and responsive remote control of robotic systems.
- PC3.** Install sensors and perform physical adjustments to ensure accurate detection and interaction with environmental elements.
- PC4.** Use manufacturer-provided control software to configure, operate, and autonomously manage the associated hardware.
- PC5.** Apply principles of wireless communication to establish stable and effective robot-to-controller connectivity.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of actuators and their hardware software integration in robotic systems.
- KU2.** Understanding of tele-operation techniques for remote control of robots.
- KU3.** Knowledge of sensor installation, calibration, and environmental interaction.
- KU4.** Knowledge of manufacturer-provided control software for system configuration and operation.
- KU5.** Knowledge of wireless communication principles for robot connectivity.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to implement and integrate actuators for effective robot interaction.

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- GS2.** Skill in applying tele-operation methods for responsive remote control.
- GS3.** Ability to install and adjust sensors for accurate detection and interaction.
- GS4.** Skill in using control software to configure and manage robotic hardware.
- GS5.** Ability to establish and maintain stable wireless communication between robot and controller.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	8	-	-
PC1. Implement and integrate actuators—both hardware and software—to enable robot interaction with the environment.	-	2	-	-
PC2. Apply tele operation techniques to achieve reliable and responsive remote control of robotic systems.	-	2	-	-
PC3. Install sensors and perform physical adjustments to ensure accurate detection and interaction with environmental elements.	-	2	-	-
PC4. Use manufacturer-provided control software to configure, operate, and autonomously manage the associated hardware.	-	1	-	-
PC5. Apply principles of wireless communication to establish stable and effective robot-to-controller connectivity.	-	1	-	-
NOS Total	-	8	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6713
NOS Name	Robot environment interaction (Autonomous Mobile Robotics)
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	1
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

Qualification Pack

ELE/N6714: Testing and fault finding (Autonomous Mobile Robotics)

Description

This NOS unit is about Testing and fault finding.

Scope

The scope covers the following :

- Analytical Techniques for Fault Finding
- Techniques and Options for Making Adjustments and Repairs
- Strategies for Problem Solving

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Apply analytical techniques to identify and diagnose faults in robotic systems.
- PC2.** Perform necessary adjustments and repairs using appropriate methods and tools.
- PC3.** Conduct test runs of individual applications and verify full system functionality, documenting all findings accurately.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of analytical techniques for fault identification and diagnosis in robotic systems.
- KU2.** Understanding of repair methods, tools, and adjustment procedures.
- KU3.** Knowledge of testing procedures, system validation, and documentation practices.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to analyse and diagnose faults in robotic systems effectively.
- GS2.** Skill in performing adjustments and repairs using appropriate tools and methods.
- GS3.** Ability to conduct test runs, verify system performance, and document results accurately.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	8	-	-
PC1. Apply analytical techniques to identify and diagnose faults in robotic systems.	-	4	-	-
PC2. Perform necessary adjustments and repairs using appropriate methods and tools.	-	2	-	-
PC3. Conduct test runs of individual applications and verify full system functionality, documenting all findings accurately.	-	2	-	-
NOS Total	-	8	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6714
NOS Name	Testing and fault finding (Autonomous Mobile Robotics)
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	1
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

Qualification Pack

ELE/N6715: Performance Review and Commissioning (Autonomous Mobile Robotics)

Description

This NOS unit is about overall performance of the robot and optimizing each system's operation through analysis, problem-solving, and refinement.

Scope

The scope covers the following :

- Test individual components of the mobile robot against agreed operating criteria.
- Optimize system operations
- Review design, fabrication, assembly, and operation processes
- Ensure all designs meet required industry standards

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Test each part of a mobile robot against agreed operating criteria.
- PC2.** Test the overall performance of a mobile robot against agreed operating criteria.
- PC3.** Apply correct criteria and methods for testing equipment and systems.
- PC4.** Conduct test runs following defined procedures and safety norms.
- PC5.** Analyse the scope and limitations of technologies used in the robot.
- PC6.** Use creative thinking strategies to generate improvements and innovations.
- PC7.** Identify possibilities for incremental and radical changes in robot performance.
- PC8.** Optimize subsystem operations through analysis, problem solving and refinement.
- PC9.** Review the entire lifecycle design, fabrication, assembly, and operation against accuracy, consistency, time and cost criteria.
- PC10.** Finalize and present technical documentation and portfolios, and respond to client queries.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Knowledge of testing criteria and performance standards for mobile robotic components and systems.
- KU2.** Understanding of testing methods, procedures, and safety norms.
- KU3.** Knowledge of system performance analysis and evaluation techniques.
- KU4.** Knowledge of innovation strategies and improvement methodologies
- KU5.** Knowledge of lifecycle stages including design, fabrication, assembly, and operation.

Generic Skills (GS)

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User/individual on the job needs to know how to:

- GS1.** Ability to test individual components and overall robot performance against defined criteria.
- GS2.** Skill in applying appropriate testing methods and following safety procedures.
- GS3.** Ability to analyse system performance and identify limitations.
- GS4.** Skill in applying creative thinking for innovation and improvements.
- GS5.** Ability to identify and implement incremental and radical performance enhancements.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	-	28	3	-
PC1. Test each part of a mobile robot against agreed operating criteria.	-	3	-	-
PC2. Test the overall performance of a mobile robot against agreed operating criteria.	-	3	-	-
PC3. Apply correct criteria and methods for testing equipment and systems.	-	3	-	-
PC4. Conduct test runs following defined procedures and safety norms.	-	3	-	-
PC5. Analyse the scope and limitations of technologies used in the robot.	-	3	-	-
PC6. Use creative thinking strategies to generate improvements and innovations.	-	3	-	-
PC7. Identify possibilities for incremental and radical changes in robot performance.	-	3	-	-
PC8. Optimize subsystem operations through analysis, problem solving and refinement.	-	3	3	-
PC9. Review the entire lifecycle design, fabrication, assembly, and operation against accuracy, consistency, time and cost criteria.	-	2	-	-
PC10. Finalize and present technical documentation and portfolios, and respond to client queries.	-	2	-	-
NOS Total	-	28	3	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ELE/N6715
NOS Name	Performance Review and Commissioning (Autonomous Mobile Robotics)
Sector	Electronics
Sub-Sector	
Occupation	Product Design & Development-EM&B
NSQF Level	4.5
Credits	5
Version	2.0
Last Reviewed Date	06/02/2026
Next Review Date	06/02/2028
NSQC Clearance Date	06/02/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ELE/N6708.Work organization and management (Autonomous Mobile Robotics)	-	10	-	-	10	10
ELE/N6709.Communication and interpersonal skills (Autonomous Mobile Robotics)	-	10	-	-	10	10
ELE/N6707.Design (Autonomous Mobile Robotics)	-	10	-	-	10	10
ELE/N6711.Navigation and localization (Autonomous Mobile Robotics)	-	8	-	-	8	8
ELE/N6710.Prototyping (Autonomous Mobile Robotics)	-	10	-	-	10	10
ELE/N6712.Vision (Autonomous Mobile Robotics)	-	8	-	-	8	8
ELE/N6713.Robot environment interaction (Autonomous Mobile Robotics)	-	8	-	-	8	8
ELE/N6714.Testing and fault finding (Autonomous Mobile Robotics)	-	8	-	-	8	8
ELE/N6715.Performance Review and Commissioning (Autonomous Mobile Robotics)	-	28	3	-	31	28
Total	0	100	3	-	103	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.